Chemical space: concept and visual (graphic) representation

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- Reymond, JL. Chemical space as a unifying theme for chemistry. J Cheminform 17, 6 (2025). https://doi.org/10.1186/s13321-025-00954-0.
- J. L. Medina-Franco, A. L. Chávez-Hernández, E. López-López, F. I. Saldívar-González, Mol. Inf. 2022, 41, 2200116.

Content

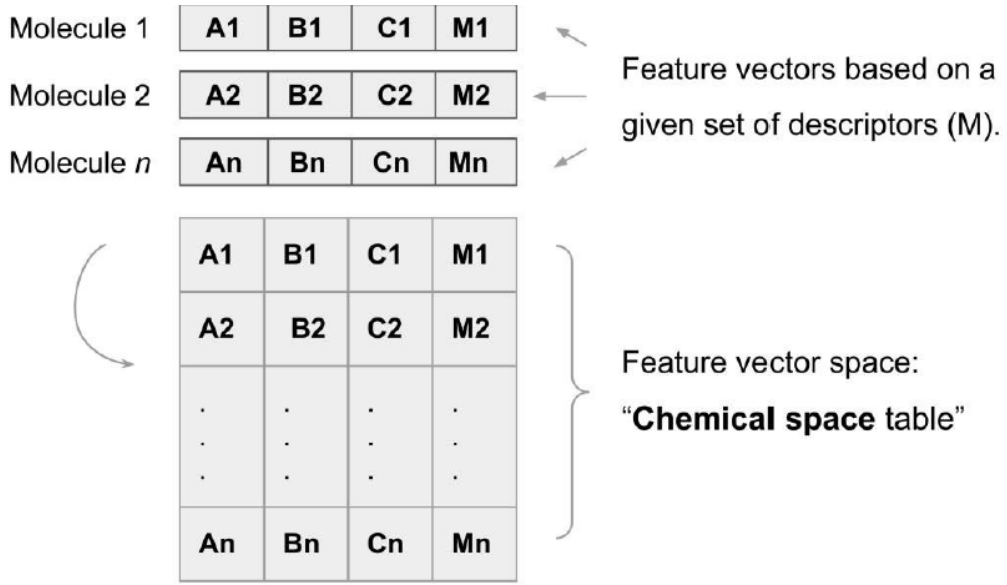
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Concepts and visual representations

- What is chemical space
- Chemical space as a unifying theme
- Visual representations and dimension reduction
- Chemical multiverse
- Chemical multiverse vs consensus chemical space
- Some practical applications

Reflections

What is chemical space?



"An M-dimensional cartesian space in which each molecule is represented as a vector of M physicochemical, topological, or pharmacological descriptors"

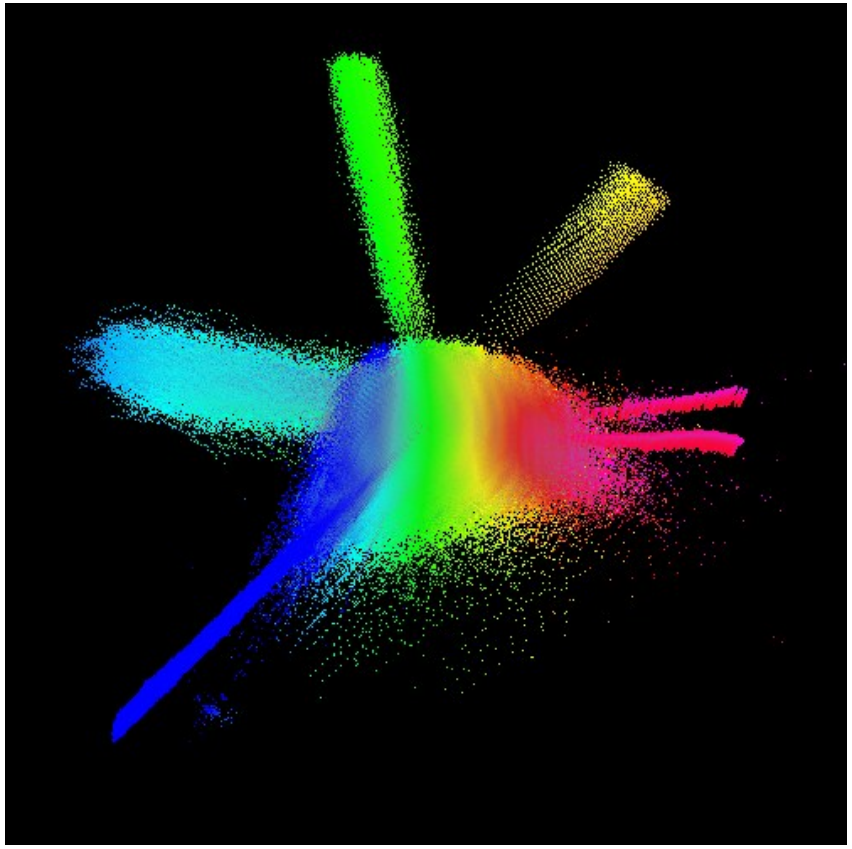
Table 2. Examples of definitions of chemical space concepts, as proposed in the literature.[a]

Author(s)	Chemical space definitions
Dobson	"All possible small organic molecules, including those present in biological systems".
Lipinski and Hopkins	"Chemical space can be viewed as being analogous to cosmological universe in its vastness, with chemical compounds populating space instead of stars".
Reymond, et al.	"Ensemble of all known and possible molecules described by their chemical properties".
Varnek and Baskin	"The ensemble of graphs or descriptor vectors forms a chemical space in which some relations between the objects must be defined".
von Lilienfeld, et al.	"The combinatorial set of all compounds that can be isolated and constructed from possible combinations and configurations of N_1 atoms and N_e electrons in real space".
Virshup et al.	"An M-dimensional cartesian space in which compounds are located by a set of M physicochemical and/or chemoinformatic descriptors".
Vogt	"Comprehensive collection of all possible small molecules under some reasonable restrictions considering size and composition".
Huang and Lilienfeld	"Chemical compound space is the set of all theoretically conceivable combinations of chemical elements and (meta-)stable geometries that make up matter".

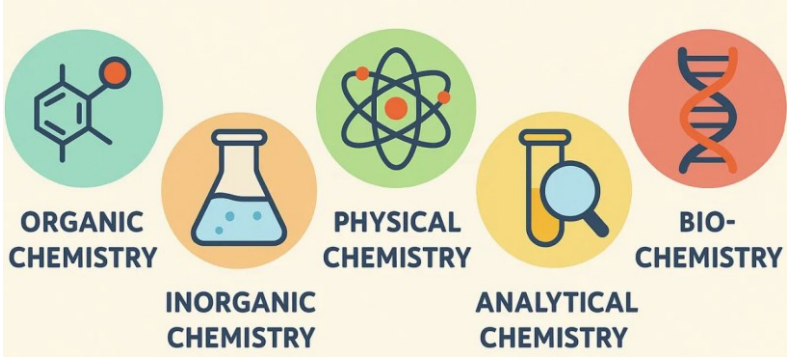
[a] In chronological order of publication.

Chemical space as a unifying theme

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It enables historically isolated subdisciplines (drug discovery, organic synthesis, asymmetric catalysis, materials science, quantum chemistry, and toxicology) to be connected through a common graphical and numerical language.

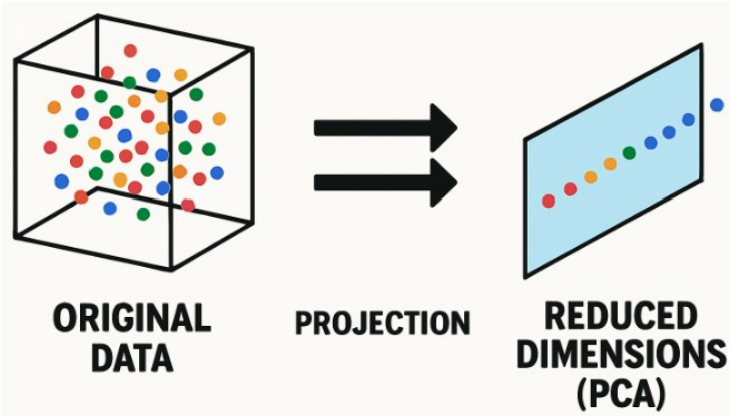


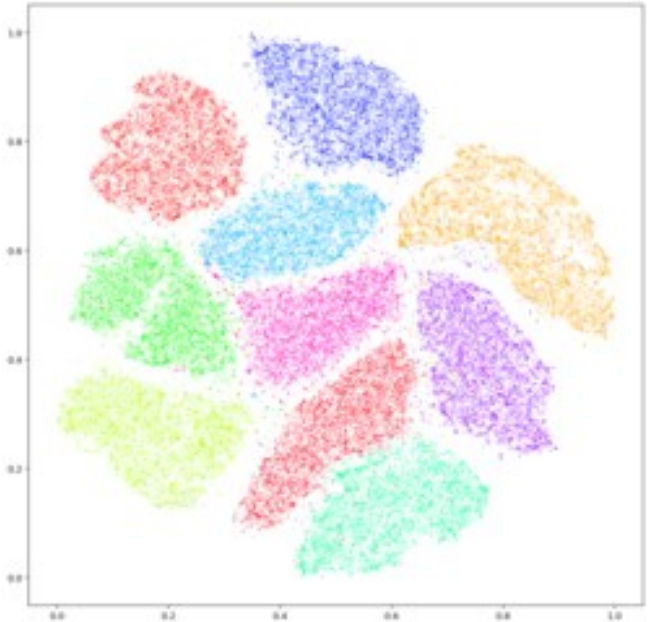
Replace blind trial and error with the intentional exploration of molecular similarity landscapes

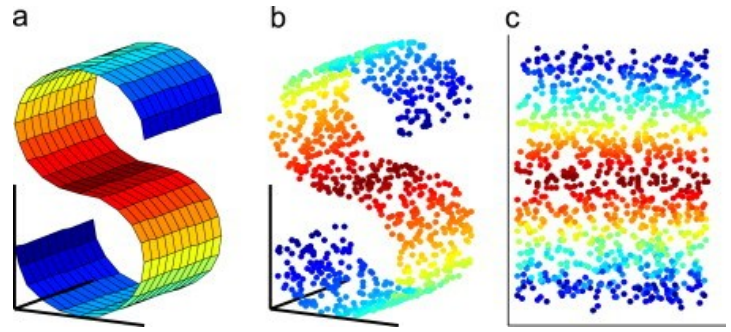
- https://gdb.unibe.ch/chemical-space-as-a-unifying-theme-for-chemistry
- https://www.workybooks.com/resources/physical-science/chemistry/

Visual representation and dimension reduction

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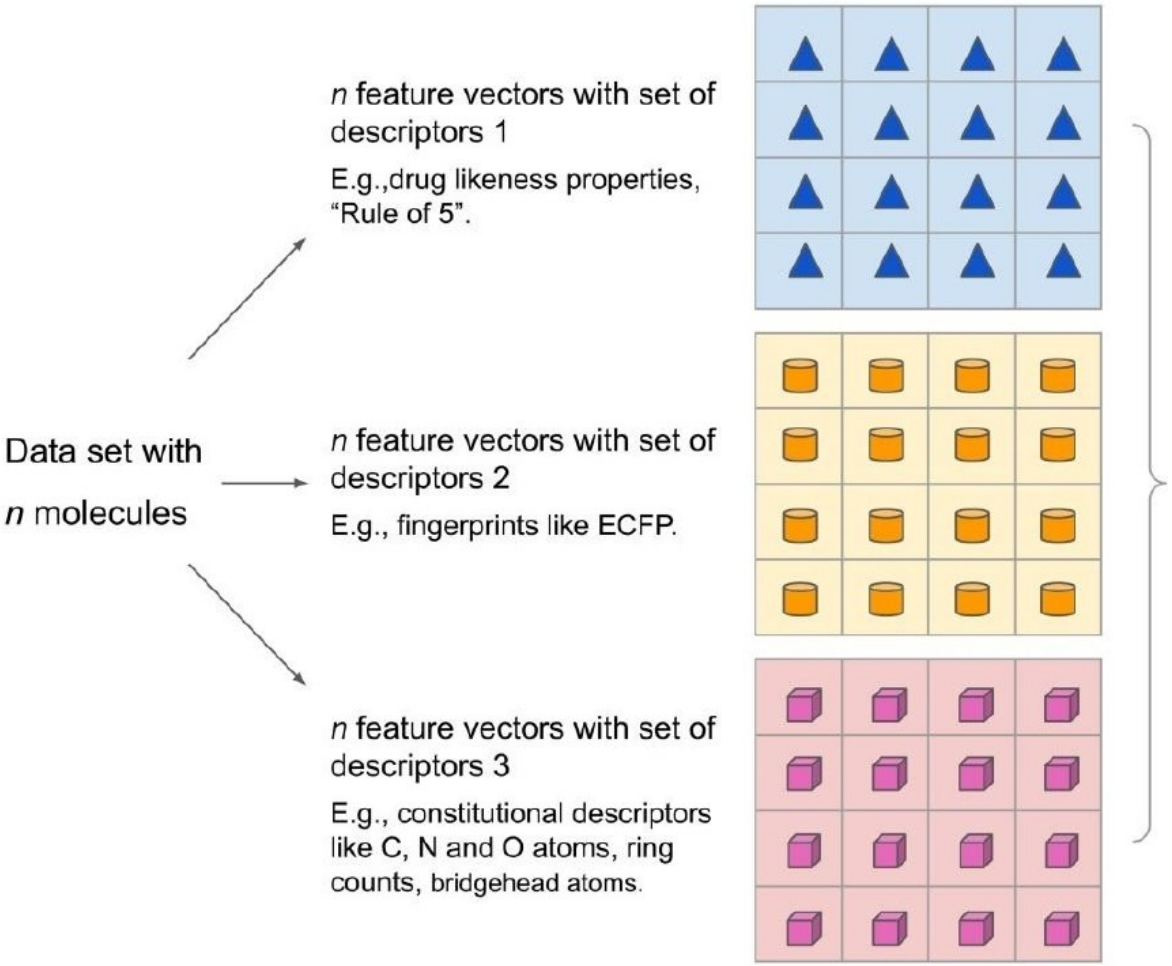
PCA

t-SNE

UMAP

- https://medium.com/@kiranvutukuri/27-principal-component-analysis-pca-dimensionality-reduction-b7ed1b724a02
- https://www.jkobject.com/blog/umap-explanation/
- https://en.wikipedia.org/wiki/T-distributed_stochastic_neighbor_embedding

Chemical multiverse

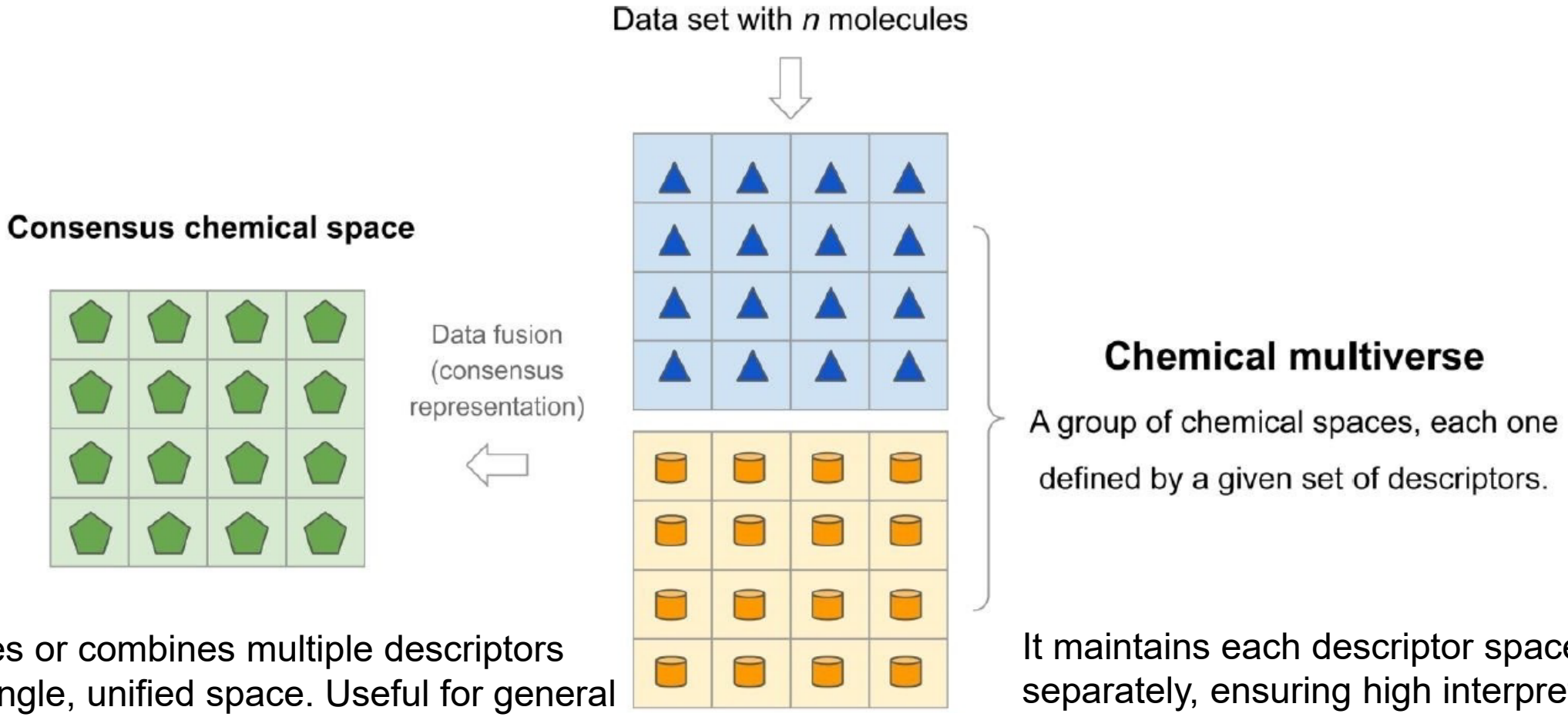


Defined as the collection of multiple parallel chemical spaces generated for the same set of n molecules using different sets of descriptors ("descriptor universes")

(Ensures that the similarity relationships specific to each representation are not lost)

Chemical multiverse vs consensus chemical space

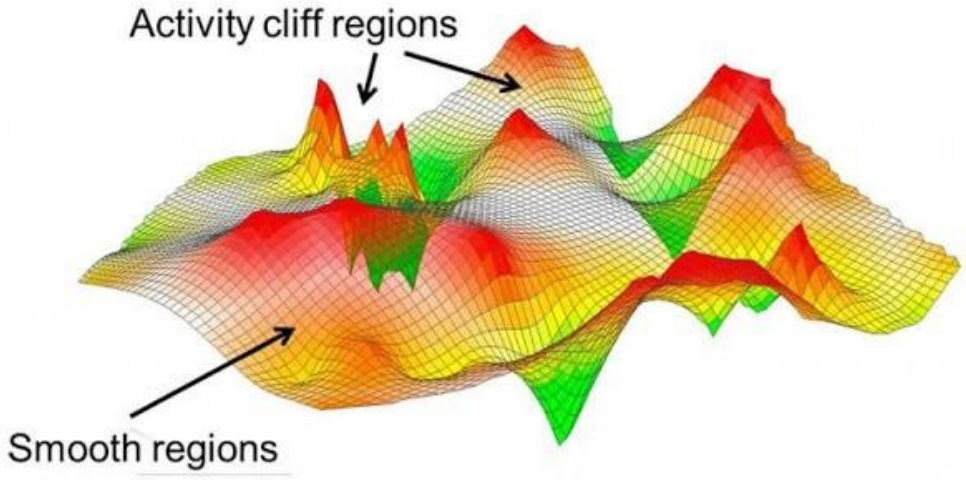
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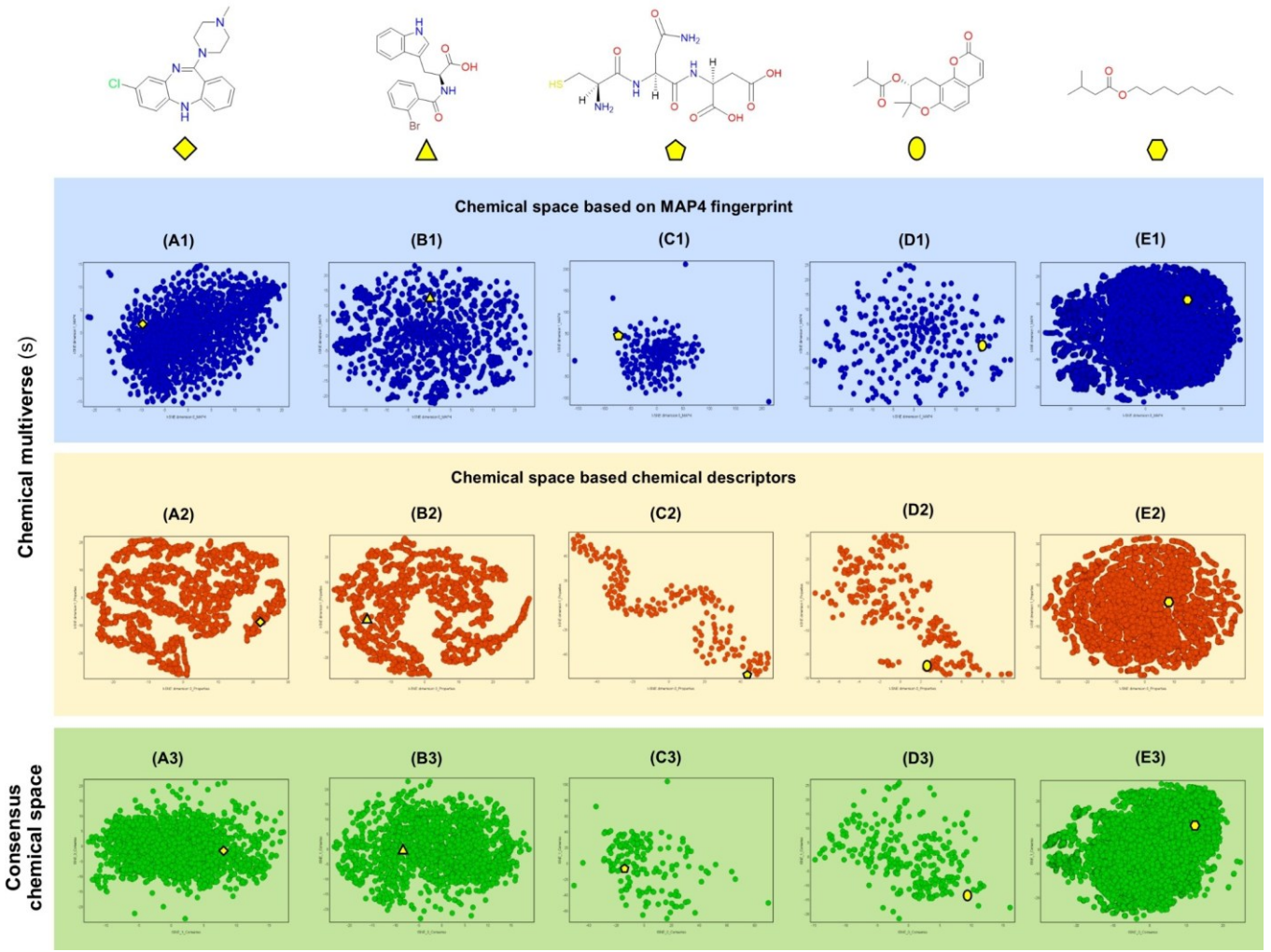
Averages or combines multiple descriptors into a single, unified space. Useful for general classifications, although it can make specific structural interpretation more difficult

It maintains each descriptor space separately, ensuring high interpretability when tracking which specific property determines the molecular position

Some practical applications



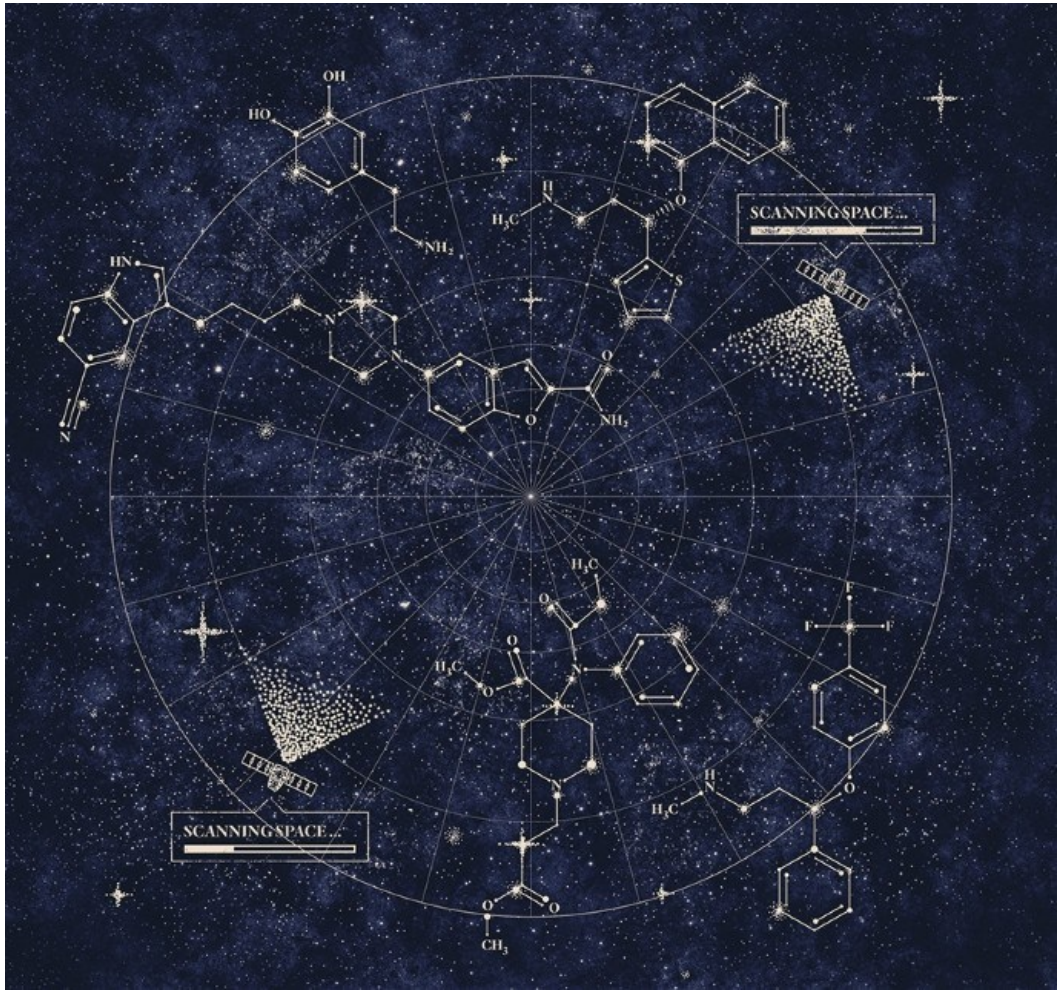
SAR



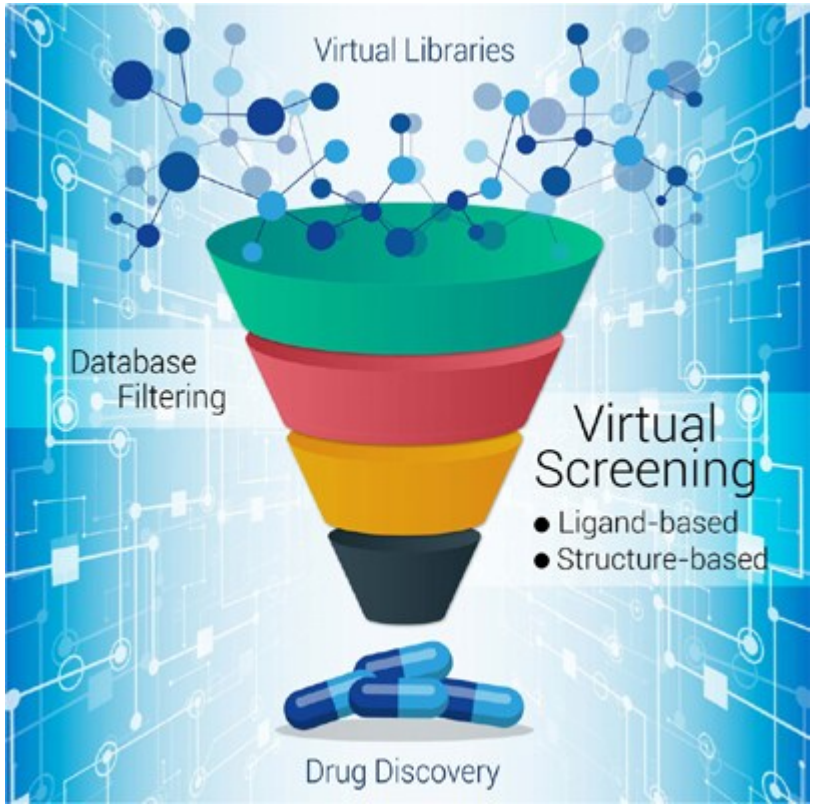
Visual representation

- http://www.int.laborundmore.com/archive/281210/Data-Visualization-in-Medicinal-Chemistry.html

Some practical applications



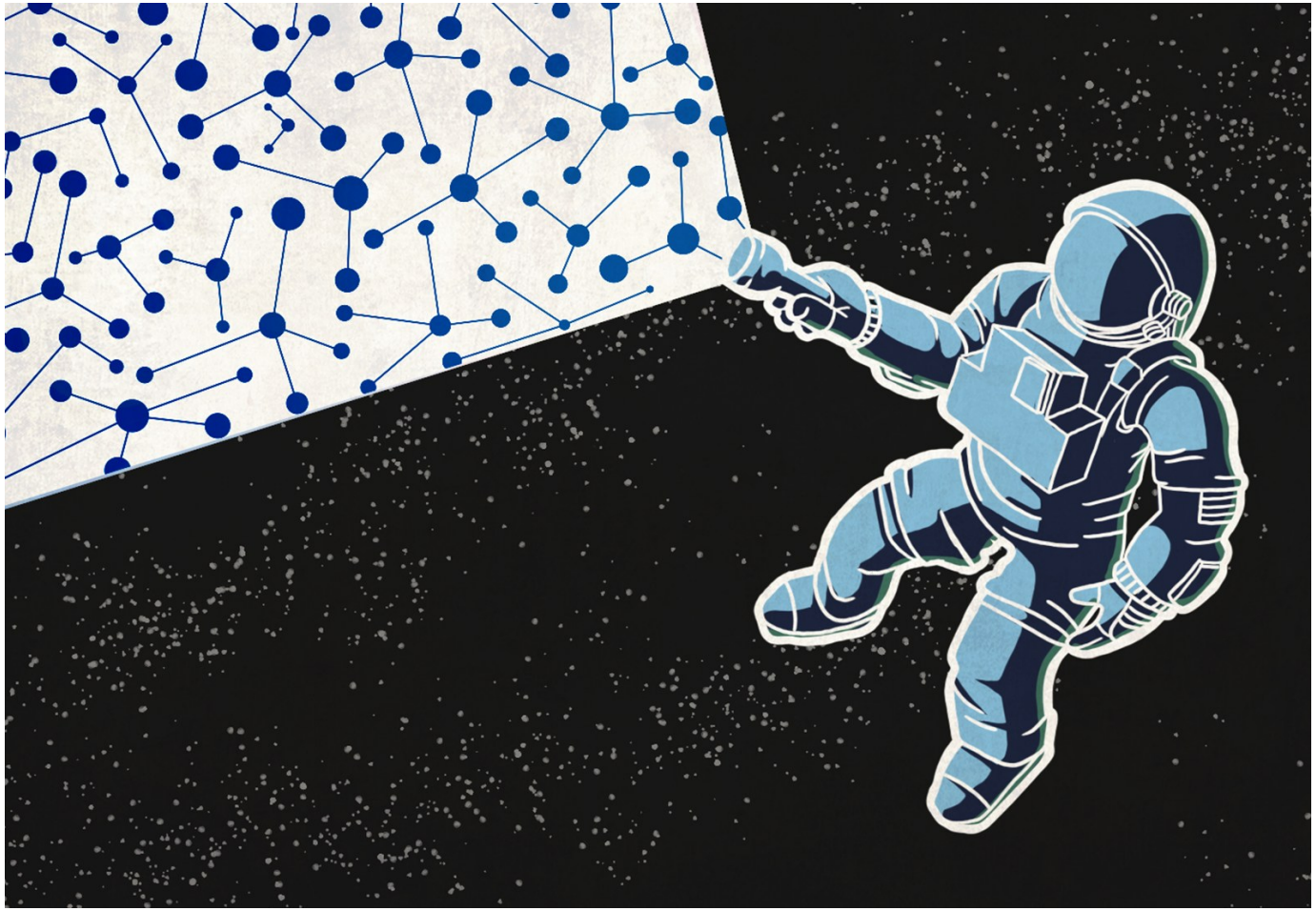
Chemical space navigation



VS

Mullard, A. The drug-maker's guide to the galaxy. Nature 549, 445–447 (2017). https://doi.org/10.1038/549445a
Virtual Screening Techniques in Drug Discovery: Review and Recent Applications. Current Topics in Medicinal Chemistry, Volume 19, Issue 19, 2019

Reflections



- Chemical space and the chemical multiverse are indispensable theoretical and graphical tools for modern computational and medicinal chemistry.
- It is not unique: it changes with the descriptors, so it pays to look at several (multiverse).
- Fusing descriptors (consensus) gives a global view but may lose information.

https://extrapolations.com/what-is-chemical-space/

90	7	19
Th	N	K
Thorium	Nitrogen	Potassium
232.03806	14.0067	39.0983
2 8 18 32 18 10 2	2 5	2 8 8 1

39	8	92
Y	O	U
Yttrium	Oxygen	Uranium
88.90585	15.9994	238.02891
2 8 18 9 2	2 6	2 8 18 32 21 9 2

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